AI BASED DIAGNOSIS SYSTEM

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Paper Name & code :-

Project & 2936810003

OBJECTIVE

- To develop an AI-based medical diagnosis system that predicts possible diseases based on user-input symptoms.
- To utilize Artificial Intelligence and Machine Learning techniques for improving diagnostic accuracy.
- To enable early detection of diseases and assist in timely decision-making.
- To provide a fast, simple, and user-friendly platform for users to check their health conditions.
- To make healthcare support more accessible and efficient, especially in areas with limited medical facilities.

Work Distribution

Member 3 → Frontend Developer

- Design UI (HTML, CSS, JS / React)
- Connect frontend with backend APIs
- Display outputs (graphs, predictions, insights)(
- Ensure responsive design (desktop)

Member 4 → Backend Developer

- Focus: Build server,API
- Connect backend with ML model
- Handle requests & responses
- Manage authentication (if needed)
- Ensure smooth communication between frontend & ML

Member 2 →Data engineer

- Collect dataset (CSV, APIs, etc.)
- Clean data
- Store data
- Data preprocessing
- Provide clean dataset to ML team

Member 1 & 5 →Feature & ML engineer

- Create new features from raw data
- Analyze feature importance
- Improve data quality for model
- Build model using LightGBM algorithms
- Train and test model
- Save model
- Provide model to backend

Data Engineering Pipeline

Raw Dataset

- Training.csv
- 246,945 rows
- 378 columns
- Symptoms + Disease

Issues:

- Duplicate rows
- Inconsistent naming

Processing Pipeline

Cleaning

- Standardized column names
- Cleaned disease labels
- Removed 57k duplicates

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Filtering

- Removed rare diseases
- Kept ≥7 samples per class

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Splitting

- Train: 70%
- Validation: 15%
- Test: 15%
- Stratified split

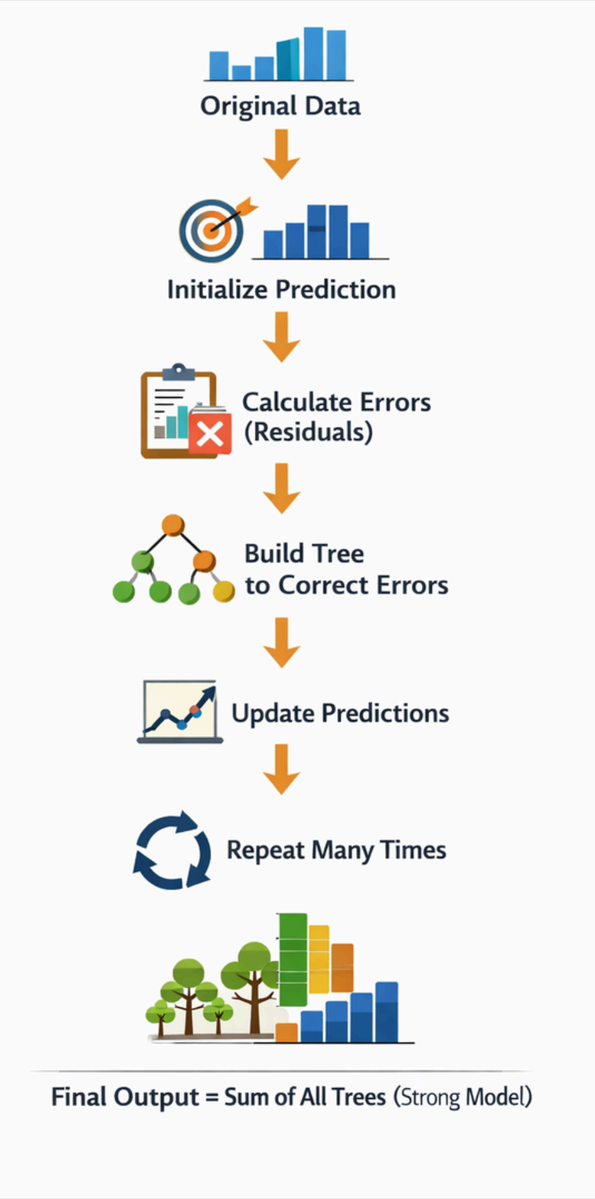
Final Data

- ~189,000 rows
- Clean & structured
- ML-ready dataset

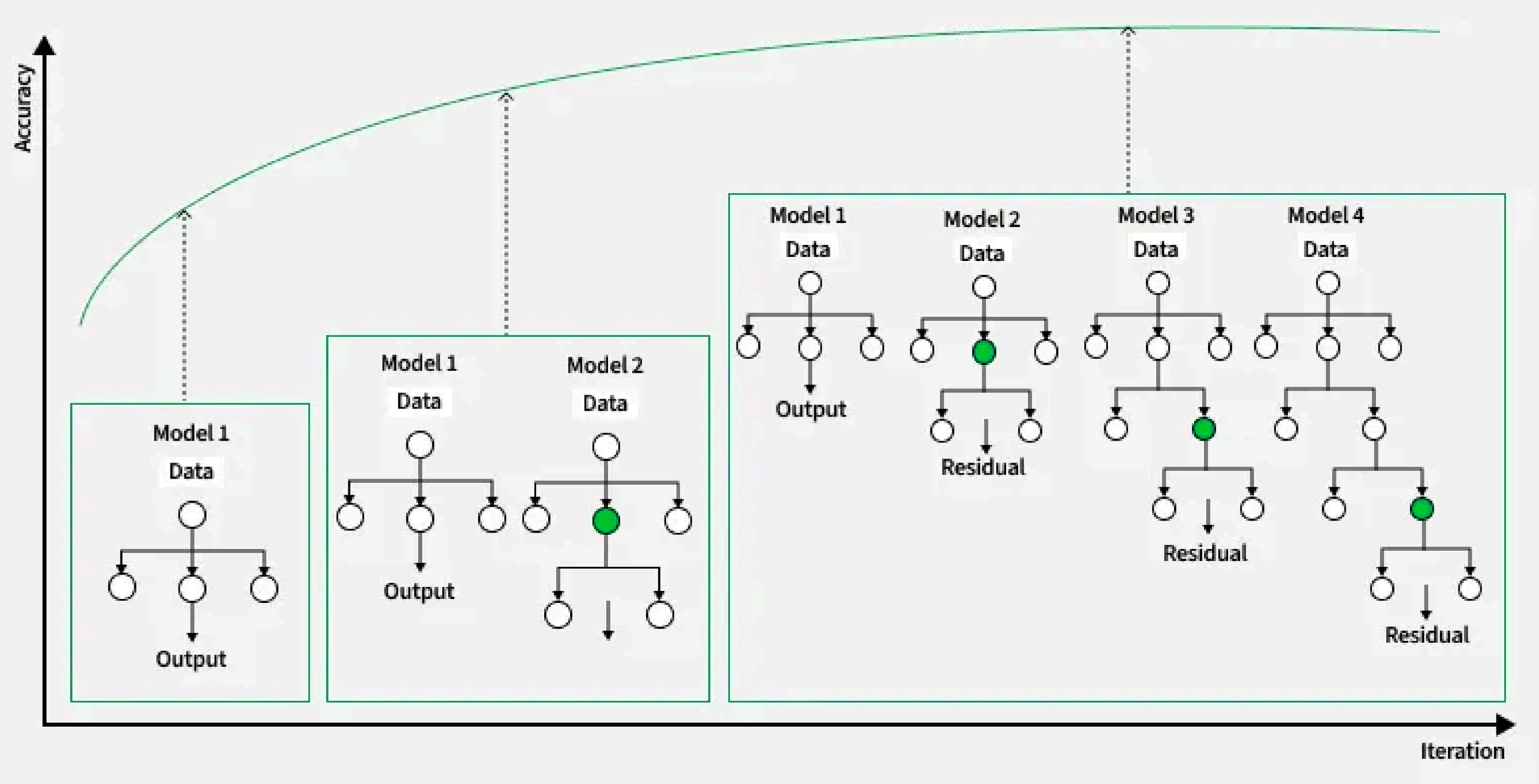
LIGHT GBM

Light GBM is a gradient boosting framework that uses tree based learning algorithms.
Get about 70% accuracy in the model by light gbm .
Steps :-

- Take the full training data and Initialize the model with a simple prediction
- Compare predicted values with actual values and Find the errors (residuals) of the model.
- Create a tree that learns to predict these errors. this tree is not random–it is focused on mistakes.
- LightGBM grows the tree leaf by leaf and It chooses the leaf that reduces error the most and splits it first.
- New prediction = old prediction + (learning rate × new tree output)
- Repeat the process and Final output=Majority probability



LIGHT GBM WORKFLOW



ADVANTAGES AND CHALLENGES

Advantages :-

- Provides fast and efficient diagnosis based on user symptoms.
- Improves accuracy using Artificial Intelligence techniques.
- Saves time compared to traditional diagnosis methods.
- Accessible from anywhere through a web-based platform.
- Supports early detection of diseases.

Challenges :-

- Ensuring accuracy and reliability of predictions.
- Handling incorrect or incomplete user inputs.
- Maintaining data privacy and security.
- Limited effectiveness for rare or complex diseases.
- Dependence on quality of training data.

THANK YOU